

Aim

To install a C compiler (GCC) in the Linux VM and execute simple programs.

Procedure

Step 1: Open Terminal

In the Ubuntu VM, press Ctrl+Alt+T to open the terminal.

Step 2: Update Package Repository

```
sudo apt update
```

Step 3: Install Build Essentials

The build-essential package includes the GCC compiler and other necessary tools.

```
sudo apt install build-essential -y
```

Step 4: Verify Installation

```
gcc --version
```

Step 5: Write a Simple C Program

Create a file named greatest.c using a text editor like nano:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a, b, c;
```

```
    printf("Enter the value of a:");
```

```
    scanf("%d", &a);
```

```
    printf("Enter the value of b:");
```

```
    scanf("%d", &b);
```

```
    printf("Enter the value of c:");
```

```
    scanf("%d", &c);
```

```
    if (a > b && a > c)
```

```
    {
```

```
        printf("A is the greatest");
```

```
    }
```

```
    else if (b > c)
```

```

    {
        printf("B is the greatest");
    }
    else
    {
        printf("C is the greatest");
    }

    return 0;
}

```

Step 6: Compile and Execute

```

gcc greatest.c -o greatest
./greatest

```

Output should display: Hello, Virtual Cloud World!

Sample Program:Greatest Check

A second program, Greatest.c, was written to check whether a given integer is a palindrome:

```

ubuntu@ubuntu: ~ — nano greatest.c
GNU nano 8.7.1 greatest.c
#include <stdio.h>
int main(){
    int a,b,c;
    printf("Enter the value of a:");
    scanf("%d",&a);
    printf("Enter the value of b:");
    scanf("%d",&b);
    printf("Enter the value of c:");
    scanf("%d",&c);
    if (a>b && a>c){
        printf("A is the greatest");
    }
    else if (b>c){
        printf("B is the greatest");
    }
    else{
        printf("C is the greatest");
    }
    return 0;
}
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify  ^_ Go To Line
Read 20 lines

```

Output

The greatest.c program was compiled and executed, and tested with two inputs (121 and 123):

```
ubuntu@ubuntu:~$ nano greatest.c
ubuntu@ubuntu:~$ gcc greatest.c -o greatest
ubuntu@ubuntu:~$ ./greatest
Enter the value of a:23
Enter the value of b:21
Enter the value of c:30
C is the greatest
ubuntu@ubuntu:~$
```

Result

Thus, the GCC compiler was successfully installed in the Linux VM and used to compile and run C programs.